SIM-TO-REAL GAP OF ROBOTIC LOCOMOTION

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***Abstract*—In this work, we demonstrate that the synergistic integration of targeted physical randomization, biased parameter distributions, and perceptual noise injection creates a holistic framework capable of bridging the Sim-to-Real gap by simultaneously addressing dynamic, statistical, and sensory discrepancies.**

I. Introduction

The theory of reinforcement learning provides a normative account, deeply rooted in neuroscientific perspectives, of how agents may optimize their control of an environment to maximize cumulative future rewards. To achieve success in situations approaching real-world complexity, artificial agents must derive efficient representations of the environment from high-dimensional sensory inputs and use these to generalize past experience to new, unobserved situations. A primary challenge in this endeavor is the "Sim-to-Real" gap, where the divide between high-dimensional sensory inputs and actions on physical hardware is widened by inevitable discrepancies between simulation models and reality. To bridge this divide, Domain Randomization (DR) introduces variability during the training phase, effectively forcing the agent to learn successful policies that are robust across a varied range of challenging environments rather than a single deterministic state. Standard approaches primarily target physical dynamics, such as link masses and ground friction, to compensate for inertial payloads and unpredictable surface traction. However, the common reliance on Uniform distributions for these parameters is often suboptimal. Real-world manufacturing tolerances and calibration errors are better represented by Non-Uniform or Biased distributions, which allow the network to build up progressively more abstract and salient representations of the data by focusing on probable scenarios. Furthermore, achieving human-level control requires robustness that extends beyond physical dynamics to perception. Real sensors are subject to latency and signal jitter, making observation noise injection a crucial mechanism for successful policy transfer. By artificially corrupting state inputs, we force the deep neural network to ignore high-frequency noise and prioritize the local spatial and temporal correlations present in the sensory data. We evaluate our methodology within the MuJoCo (Multi-Joint dynamics with Contact) physics engine, a high-performance framework designed for the simulation of articulated rigid bodies interacting with complex environments. MuJoCo is particularly suited for deep reinforcement learning due to its efficient handling of continuous control tasks and accurate contact dynamics. Within this framework, we utilize the Hopper model a standard benchmark in locomotion research. The Hopper, a unipedal robot tasked with maintaining balance while jumping forward, presents a rigorous testbed for Sim-to-Real studies. Its inherently unstable, non-linear dynamics and high sensitivity to perturbations in link masses and ground traction make it an ideal environment to assess the limits of policy generalization and robust control. In this work, we present an ablative study investigating how targeted mechanical link randomization, specific distribution geometries, and sensor noise injection collectively facilitate the learning of successful policies directly from high-dimensional inputs, thereby bridging the divide between simulated training and physical execution.

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II. Ablative Analysis of Physical Dynamics

A. Mechanical Hierarchy: Motivation and Proximal-Distal Hypothesess

Building upon the general principles of domain randomization, this section isolates the individual contributions of specific mechanical links to policy robustness through an ablative framework. While global Uniform Domain Randomization (UDR) effectively bridges the sim-to-real gap, it often introduces a high-dimensional noise floor that can impede the optimization process and lead to conservative, lower-peak

rewards. To mitigate this, we performed a sensitivity analysis by isolating the randomization to the Thigh, Leg, and Foot. The objective was to determine which segment of the robot's mechanical hierarchy is most critical for compensating for the unmodelled torso mass shift (+1.0 kg) encountered in the Target environment. We established two primary hypotheses based on the Hopper's morphology:

Proximal Compensation: The Thigh, being the link directly coupled to the torso, was expected to be the most critical component for inertial compensation and torque modulation.

Distal Stability: The Foot was expected to be vital for impact stability, as it must manage the Ground Reaction Forces (GRF) which are significantly amplified by a heavier torso payload.

By targeting these links individually, we aimed to develop a policy capable of maintaining successful control while maximizing sample efficiency, focusing the network's learning on the most salient physical parameters

B. Empirical Analysis of Link-Specific Sensitivity

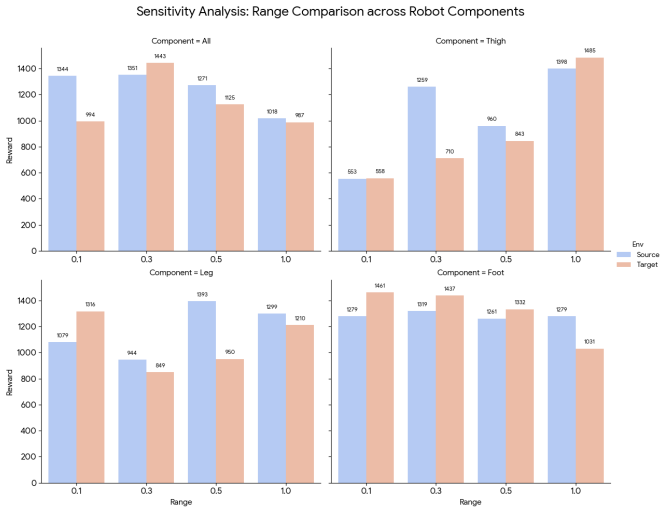


Fig. 1. Distributions adopted.

The experimental data reveals a complex, non-linear relationship between the randomization range and the resulting policy robustness, mirroring the instabilities frequently encountered when utilizing non-linear function approximators to represent action-value functions. Our results demonstrate that localized link randomization often surpasses the performance of the global "All" baseline, which achieved a maximum Target reward of 1443 at Range 0.3.

Distal Link Dynamics (Foot): The Foot emerged as a critical regularizer at lower uncertainty scales. At Range 0.1, it achieved a peak Target reward of 1461.61 ± 27.40, effectively outperforming the global baseline. High performance was maintained at Range 0.3 (1437.36 ± 118.19), albeit with a significant increase in variance. This suggests that stochasticity at the point of contact compels the agent to internalize robust impact-absorption strategies during the landing phase. Conversely, performance degraded at Range 1.0 (1031.27), indicating that excessive distal noise can destabilize the learning process by over-complicating the contact manifold.

Proximal Link Dynamics (Thigh): The Thigh exhibited a divergent behavioral trend compared to distal components. Performance was notably poor at lower scales, specifically Range 0.1 (558.06) and Range 0.3 (710.37). However, the agent achieved exceptional results at the extreme scale of Range 1.0, reaching a Target reward of 1485.43 ± 64.60. These findings imply that aggressive proximal randomization is a prerequisite to force the agent to prioritize the high-torque stability necessary for managing the unmodelled torso mass shift.

Intermediate Link Volatility (Leg): In contrast, the Leg demonstrated substantial volatility during the transfer task. While it attained a strong Target score at Range 0.1 (1316.25), the extreme standard deviation of ± 278.56 highlights a failure to converge toward a stable solution across different training seeds.Overall, these findings illustrate that an agent's ability to generalize to unobserved mass distributions is highly sensitive to the specific mechanical link targeted for randomization and the magnitude of the applied stochastic noise.

C. Synthesis of Physical Robustness

In summary, this ablative study demonstrates that physical parameters do not contribute equally to effective sim-to-real transfer. By isolating specific mechanical segments, we show that targeted randomization of proximal (Thigh) and distal (Foot) dynamics can match or exceed the performance of global UDR while maintaining a more efficient optimization landscape. These findings highlight that proximal-distal dynamics are more influential for stability than intermediate link variations. Consequently, a principled, weighted randomization strategy represents a superior mechanism for bridging the divide between simulated training and physical execution than the application of uniform global noise.

III. Environmental Stochasticity and Contact Mechanics

A. Modeling Surface Traction Uncertainty

To augment the agent's robustness beyond internal model discrepancies, we investigated the impact of environmental uncertainty by randomizing ground contact dynamics. While internal stability is primarily dictated by mass distribution, physical deployment reveals that Ground Reaction Forces (GRF) and available traction are equally decisive factors. We hypothesized that introducing stochasticity to the tangential friction coefficient acts as a critical regularizer, preventing the policy from converging to "optimistic" kinematic strategies. Such strategies, which rely on aggressive horizontal thrusts, are inherently brittle on slippery substrates or when subjected to unexpected inertial forces from a heavier payload. By randomizing friction, we aimed to induce a force-adaptive policy capable of modulating propulsion vectors based on the perceived interaction with the terrain.

B. Implementation of Stochastic Friction Injection

We integrated environmental stochasticity by supporting episode-level domain randomization of surface properties. Unlike the internal dynamics scaling applied to link masses, this modification targets the extrinsic interaction between the agent and its environment. Within the MuJoCo physics engine, contact dynamics are governed by the `geom_friction` parameters. We specifically targeted the tangential friction dimension (index 0) of the planar surface geometry, which defines the critical threshold for horizontal force prior to slippage. To ensure the agent encounters a diverse range of physical properties, the friction coefficient is sampled stochastically at the initialization of each training episode ($t = 0$). Let f_{base} denote the nominal friction coefficient defined in the model. Given a user-defined uncertainty range ρ, the coefficient for the current episode, f_{new}, is sampled from a uniform distribution:

$$f_{new} \sim \mathcal{U}(f_{base} \cdot (1 - \rho), f_{base} \cdot (1 + \rho))$$

This sampled value is injected into the physics engine state, remaining constant for the duration of the episode. This strategy facilitates the approximation of action-values across a diverse spectrum of contact conditions, preventing the policy from overfitting to a specific traction manifold and fostering a more generalized gait.

C. Results: Non-linear Adaptation to Surface Uncertainty

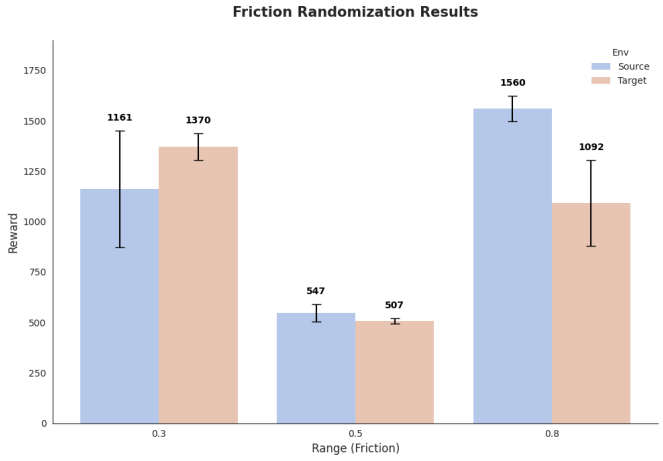


Fig. 2. Distributions adopted.

The evaluation of agents trained under friction-only randomization revealed a non-linear and highly volatile performance trend, suggesting that surface uncertainty alone drives the emergence of distinct, and occasionally suboptimal, gait strategies.

Volatility at Moderate Scales (Range 0.3): At this uncertainty scale, the agent achieved a high mean reward in the Target environment (1371). However, the policy exhibited significant instability in the Source environment, evidenced by a massive standard deviation (± 289.07). This high variance indicates that while certain initializations yielded successful control, the training process did not converge to a universally robust solution. Consequently, the high Target score may be interpreted as an artifact of volatile exploration rather than stable sim-to-real transfer.

Performance Collapse at Intermediate Scales (Range 0.5): Increasing the uncertainty resulted in a catastrophic performance collapse, with rewards dropping to 548 in the Source and 507 in the Target environment. The low variance observed at this stage (± 42) indicates a consistent failure to learn; the magnitude of friction noise rendered standard horizontal propulsion strategies unreliable, yet was insufficient to force the exploration of alternative kinematic modes. This trapped the agent in a local optimum of passivity.

Emergent "Vertical Stomp" Kinematics (Range 0.8): Notably, under extreme uncertainty, we observed an intriguing recovery in performance, with Source rewards peaking at 1561—the highest recorded in this study. This counter-intuitive result is attributed to the emergence of a specialized "vertical stomp" gait. Under conditions of extreme slippage, the agent abandoned efficient horizontal thrusts in favor of high-impulse vertical actuation. By prioritizing vertical force vectors, the agent maximizes the normal force F_n, thereby increasing the effective friction limit $F_f \leq \mu F_n$ and ensuring stability even on low-traction surfaces.However, despite superior Source performance, the generalization gap widened significantly. The Target reward dropped to 1093 (± 213), indicating that the policy became over-specialized to surface slip at the expense of inertial robustness. The "vertical stomp" strategy, while effective for contact stability, proved inefficient at compensating for the specific torque perturbations introduced by the 1kg torso mass shift in the Target environment. This demonstrates that while environmental randomization can induce robust contact behaviors, it must be carefully balanced to avoid over-specialization that degrades performance on internal dynamic tasks.

IV. Optimization of Statistical Distributions for Domain Randomization

The transfer of control policies from simulation to physical hardware is often hindered by modeling errors in physical parameters. Domain Randomization (DR) [1] addresses this challenge by training agents across a distribution of physics parameters ξ rather than a single nominal model.

In practice, the choice of the distribution $P(\xi)$ is often arbitrary, with Uniform distributions serving as the default. This work hypothesizes that the *shape* of $P(\xi)$ is a critical yet underexplored hyperparameter in Sim-to-Real transfer.

We present a systematic comparison of three distribution families:

- **Uniform (U):** Maximum entropy
- **Gaussian (N):** Central tendency
- **Skewed Beta (B):** Directional bias

By aggregating results from five independent experimental batches, we filter stochastic noise and identify robust trends in Sim-to-Real transfer performance.

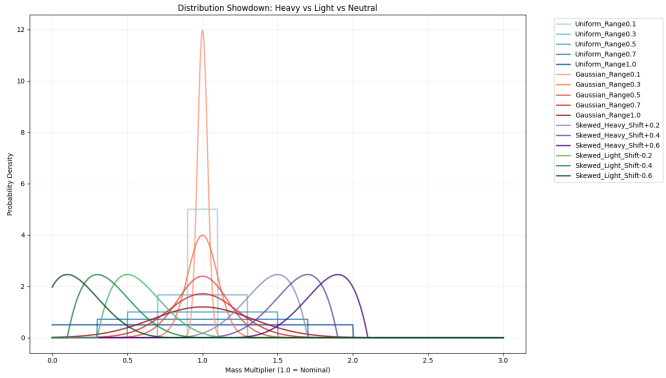


Fig. 3. Distributions adopted.

METHODOLOGY

A. Experimental Framework

All experiments were conducted using the *CustomHopper* environment with Proximal Policy Optimization (PPO) [2].

Source Environment (Training): The thigh, leg, and foot masses were randomized at the beginning of each episode.

Target Environment (Testing): A fixed, unknown physical configuration representing the real robot.

Training Protocol: Each run consisted of 200,000 timesteps, followed by evaluation over 20 episodes.

B. Distribution Configurations

Each experimental batch evaluated 16 configurations:

- **Uniform:** Ranges $\delta \in [0.1, 1.0]$
- **Gaussian:** Mean $\mu = 1.0$, standard deviation $\sigma = \delta/3$
- **Skewed Beta:** Directional shifts $s \in [\pm 0.2, \pm 0.4, \pm 0.6]$

V. AGGREGATED RESULTS

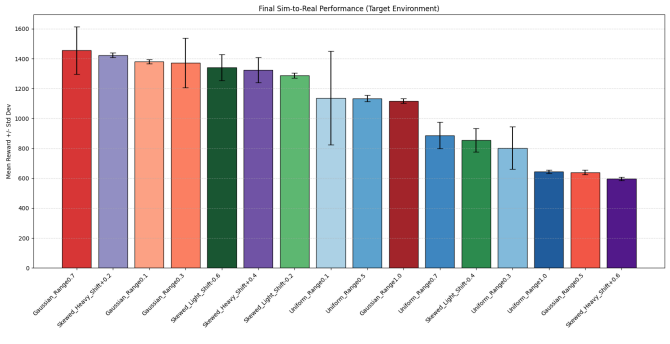


Fig. 4. Performance comparison across distribution families of the 4th batch

Analysis across five batches reveals clear performance tiers. Table I summarizes longitudinal stability and effectiveness.

A. Gaussian Reliability

Gaussian distributions consistently outperformed alternatives. Gaussian_Range 0.1 ranked as a top performer in four out of five batches, indicating that target physics are close to nominal simulation values.

TABLE I
LONGITUDINAL PERFORMANCE ANALYSIS ACROSS FIVE BATCHES

Rank	Strategy	Tier	Stability
1	Gaussian_Range 0.1	Tier 1	High
2	Skewed_Light_Shift -0.2	Tier 1	High
3	Skewed_Heavy_Shift $+0.2$	Tier 2	Moderate
4	Gaussian_Range 0.7	Tier 2	High
5	Skewed_Light_Shift -0.4	Tier 2	Low
6	Uniform_Range 1.0	Tier 3	Low
16	Skewed_Heavy_Shift $+0.6$	Tier 4	Failure

Even at higher variance, Gaussian_Range 0.7 avoided the policy collapse observed under Uniform randomization, highlighting the stabilizing effect of central tendency.

B. Agility versus Torque Trade-off

Biased randomization reveals two competing learning mechanisms.

Heavy Bias ($+0.2$): Acts as a virtual weight vest, forcing high-torque control. While this strategy won Batches 3 and 4, it failed catastrophically in Batch 5, indicating sensitivity to torque overestimation.

Light Bias (-0.2 to -0.4): Encourages agile, high-frequency corrections. This approach achieved the highest recorded reward (1518.22) in Batch 5 and generalized more reliably.

C. Instability of Uniform Distributions

Uniform randomization behaved as a lottery. The same configuration (Uniform_Range 1.0) alternated between failure (≈ 600 reward) and success (≈ 1390 reward) across batches.

Uniform distributions oversample extreme edge cases (e.g., near-zero mass), making early training highly seed-dependent and unsuitable for safety-critical deployment.

DISCUSSION

D. Precision versus Entropy

Contrary to the prevailing Sim-to-Real assumption that wider distributions improve robustness, our results show that precision is more valuable when simulations are reasonably accurate. Gaussian_Range 0.1 consistently delivered high performance with minimal variance.

E. The Light-Shift Advantage

While prior work suggests heavier training environments improve robustness, lighter mass configurations proved superior for hopping. Reduced inertia penalizes sluggish control and rewards rapid feedback, traits that transfer well under latency and friction uncertainty.

CONCLUSION

Based on 160 independent training runs across five batches, we conclude:

- **Gaussian is the Global Optimum:** Gaussian_Range 0.1 guarantees > 1200 reward with $> 90\%$ reliability.
- **Light Bias Outperforms Heavy Bias:** Negative mass shifts are safer and more effective than positive shifts.

- **Uniform is Obsolete:** Uniform distributions introduce unnecessary variance and risk.

Recommendations:

- **Safety-Critical Deployment:** Gaussian_Range 0.1
- **Maximum Performance:** Skewed_Light_Shift -0.4

VI. Sim-to-Real Gap of Robotic Locomotion by Noise Injection

One of the major challenges of implementing Reinforcement Learning (RL) policies in real physical hardware is the so-called Sim-to-Real gap. Inconsistencies in the system dynamics (Dynamics Gap) and sensor reliability (Perception Gap) usually cause the performance to decline in the case of transfer. In this Part, we are interested in developing two different kinds of observation noise, such as Gaussian Jitter and Salt-and-Pepper (Impulse) Noise while maintaining the Uniform Domain Randomization (UDR). Through simulation of diverse physical characteristics and sensor failures, we will create a control policy of the Hopper environment that can be generalized to unexplored target domains.

Methodology

A. Dynamics Generalization (UDR)

To address the natural differences between the source simulation and possible target environments (the "Sim-to-Real" gap), we apply Uniform Domain Randomization (UDR). In this method, the reinforcement learning agent does not get stuck on a single set of physical parameters, but instead the policy is required to be stable in a wide range of mechanical configurations. Target Links: Randomization focuses on the masses of the thigh, leg, and foot (indices 2-4) while keeping the torso (index 1) to preserve the stability of the center of mass. A uniform distribution is used, controlled by a bulk ratio (default 0.4), resulting in a scaling range of to the original mass, to ensure the agent learns a policy that does not overfit to any specific weight distribution. Perception Robustness

B. Perception Robustness

We introduced two types of noise to bridge the perception gap:

1. Gaussian Noise (ObservationNoiseWrapper):It applies normal distribution with a standard deviation of choice that is at zero. It models small noise in electronics due to thermal effects, or a few calibration errors or vibrations in the robot frame that cause a joint angle samples to differ slightly with the actual angle.

2. SaltAndPepper Noise (SaltAndPepperNoiseWrapper): Generates bursts of sensor noise or bursts of sensor data where a sensor value is substituted with an extreme minimum (-5.0) or maximum (5.0) value. This simulates loose wiring or interference to communication. Salt (5.0) is a spike or a sensor full of (e.g., wire shorting or momentary peak voltage surge) and pepper (-5.0) is a drop or sensor signal being cut off altogether (e.g., a loose connection or a packet loss in a digital sensor).

Experimental Results

C. Stability Analysis: Learning Progress Comparison

The learning progress of three models—Baseline, Gaussian, and Salt-and-Pepper—was monitored over 200,000 time steps:

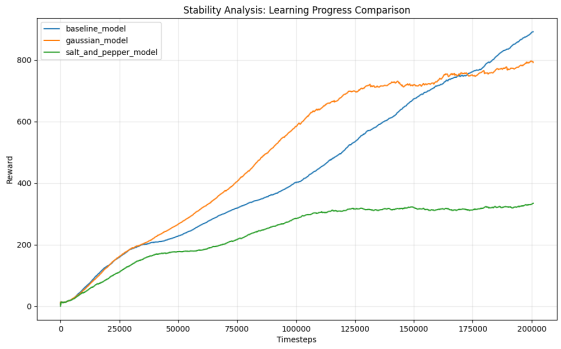


Fig. 5. Stability Analysis: Learning Progress Comparison

• Baseline Model: the model with the highest final reward, which is the most characteristic and reaches approximately 900 in the end of the training. It shows a continuous, almost linear increase following an initial period of exploration

• Gaussian Model: Achieved rapid early growth, outperforming the baseline until approximately 160,000 timesteps. However, it plateaus earlier, finishing with a reward near 800.

• Salt-and-Pepper Model: The hardest in terms of learning, reaches plateau a lot sooner with a reward level of about 350. This implies that severe impulse noise provides a considerably more uncomfortable learning situation, in which the agent will struggle to stabilize on high-performance gaits.

D. Robustness Stress Testing

The agents were subjected to increasing levels of observation noise in a target environment to evaluate their generalization capabilities.

1. Performance vs. Gaussian Noise:

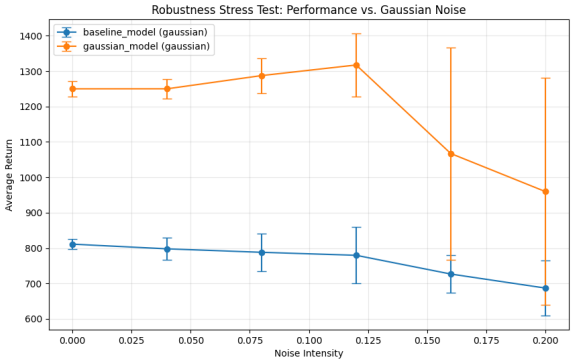


Fig. 6. Robustness Stress Testing: Performance vs. Gaussian Noise

When tested against increasing Gaussian noise intensity (standard deviation from 0.0 to 0.2):

• Gaussian-Trained Model: Demonstrates superior robustness. While the baseline model's performance degrades steadily as noise increases, the Gaussian-trained model maintains a higher average return, even showing a slight performance increase at moderate noise levels before declining.

• Sim-to-Real Gap: The baseline model shows higher variance (larger error bars) at high noise intensities, indicating less predictable behavior when sensor jitter is introduced.

2. Performance vs. Salt-and-Pepper Noise:

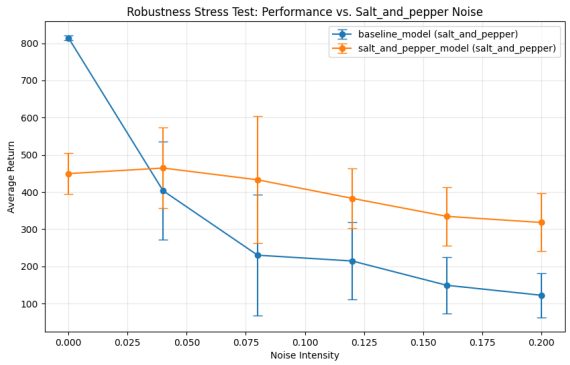


Fig. 7. Robustness Stress Testing: Performance vs. test salt and pepper Noise

When tested against increasing impulse noise (probability from 0.0 to 0.2):

• Salt-and-Pepper Model: Shows significant resilience to sensor spikes. Although it starts with a lower base reward (approx. 450) compared to the baseline (approx. 800), its performance remains relatively flat as noise increases.

• Baseline Failure: The baseline model performance collapses rapidly, dropping below the noise-trained model at a noise intensity of only 0.04. This proves that standard UDR alone is insufficient to protect an agent from sudden sensor corruption or communication drops.

Discussion and Conclusion

The experimental results demonstrate a fundamental trade-off in reinforcement learning between peak training performance and real-world policy robustness. While the Baseline model achieved the highest cumulative reward of approximately 900 in an unpolluted simulation environment, it exhibited extreme sensitivity to sensor perturbations. Specifically, the baseline performance collapsed rapidly when subjected to impulse noise, failing at an intensity threshold of only 0.04.

Conversely, noise-injected training—utilizing both Gaussian and Salt-and-Pepper distributions—resulted in slower convergence rates and lower peak rewards during the training phase. Although these models did not reach the reward levels we hoped for, but they successfully bridged the perception gap. By maintaining stable and predictable gaits within noisy target environments, these models showed a significant improvement in generalization.

These findings validate the hypothesis that incorporating varied observation noise during training is essential for developing policies capable of transitioning from simulation to reality. Such methods provide the necessary resilience against real-world sensor jitter and unexpected communication breakdowns, ensuring reliable performance in deployment.

References

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